

Shounak Shastri

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SUMMARY

Applied Scientist with a Ph.D. and nearly 5 years of industry experience shipping Computer Vision and Natural Language Processing (NLP) products for global clients at scale. End-to-end ownership from research to deployment, with hands-on experience leading cross-functional teams and managing stakeholders across India, Japan and USA. Research foundation in Steganography, with 7 peer-reviewed publications. Key Outcomes: 23% productivity gain (2 FTE saved), 81% reduction in compliance violations and 80% reduction in document processing times.

EXPERIENCE

Recode Solutions Pvt. Ltd. (<https://www.recodesolutions.com/>)

Chennai, TN, IN

Technical Analyst – Data and Applied Science (Promoted from Sr. Software Engineer)

May 2021 – Present

End-to-end ownership of applied ML projects across computer vision and NLP, serving major manufacturing clients in India, Japan and USA.

• LLM Finetuning for structured data extraction:

- Designed and owned an iterative fine-tuning pipeline for LLaMA 3.2–3B (Hugging Face Transformers + Unsloth) to extract structured JSON from 3M+ scanned chemical analysis documents for a global paint manufacturer in America, deployed fully on-premise on an NVIDIA A1000 GPU.
- Bootstrapped annotation with active learning: manually labeled 50 seed documents, used the stage-1 model to pre-annotate 1,000 more (reviewed by human annotators), then scaled to 2,500 documents — reducing annotation cost while maintaining quality.
- Defined evaluation framework using exact-match accuracy and field-level F1 scores, supplemented by human spot-checks; implemented a dashboard-based user-edit tracker to surface failure cases and drive future retraining.
- Implemented PaddleOCR-based ingestion for scanned PDFs across multi-vendor document formats with varying schemas; designed JSON output structure in collaboration with database and UI teams.
- Benchmarked supervised fine-tuned model against prompt-engineering baseline, demonstrating superior structured extraction accuracy; implemented low-temperature inference for output consistency and a human-in-the-loop edit-tracking system to collect human feedback signals for future retraining datasets.
- Reduced document processing latency by 80% compared to manual workflows and contributed to patent documentation for the automated extraction architecture.

• Computer Vision & Action Recognition for Manufacturing Productivity Analytics

- Led full-cycle development and multi-site deployment of a real-time action recognition system across 20 production-line stations for a leading Indian 2-wheeler manufacturer (3 plants) and a leading Indian passenger vehicle manufacturer (14 stations, 2 shifts); additionally deployed a warehouse efficiency monitoring system for a Japanese automotive client (25 stations) processing 16 hours of footage nightly.
- Fine-tuned EfficientNet-based video classifiers using NVIDIA TAO Toolkit for upto 15 action classes per station; mitigated class imbalance via augmentation and sample-capping; optimized inference pipeline with ONNX (internal testing) and TensorRT (deployment) on NVIDIA A1000 class GPUs.
- Designed VA/NVA classification logic in collaboration with client operations teams; built shift-wise dashboards covering cycle analysis, deployment adherence, REBA/RULA ergonomics scores, and operator wise Yamazumi charts.
- Built the company-standard annotation and data-cleaning tool adopted across all productivity deployments; trained 2 junior AI engineers on the full ML pipeline; managed end-to-end client engagements across India and Japan; led 5 additional POCs for clients in India, Australia, and Dubai.
- Achieved 23% productivity improvement and 2 FTE saved at the Indian 2-wheeler manufacturer; 100% error elimination across 7 stations. Contributed to patent documentation for the action recognition architecture.

• Object Detection for Safety and Compliance Monitoring

- Trained and deployed a YOLOv4 object detection model to monitor PPE compliance (helmets, goggles, gloves, vests) and MHE presence (forklifts, trolleys) across the CCTV network of a Japanese automotive client's warehouses and factories in real time.
- Set up an end-to-end annotation pipeline using CVAT on an internal server; instructed and managed a team of junior annotators; selected YOLOv4 for its inference speed and favorable licensing conditions.
- Optimized detection confidence thresholds per object class to minimize false positives in cluttered warehouse environments with variable lighting; balanced precision-recall tradeoffs to ensure alarm fatigue did not undermine operator trust in the system.
- Delivered an 81% reduction in overall compliance violations; built a live violation-reporting dashboard integrating real-time model output.

EDUCATION

Doctorate (Ph.D.) in Electronics and Communication Engineering Jun 2015 – Dec 2020
Vellore Institute of Technology *Vellore, TN, India*

- *Research:* Published 7 peer-reviewed papers on Steganography Algorithms and Secure Communication; received 2 research awards during the program. *ORCID Link:* <https://orcid.org/0000-0002-2247-9046>
- *Teaching & Mentoring:* Taught Image Processing and Signal Processing to 60+ students and mentored 30+ undergraduates, leading to conference presentations and research prototypes.

Master of Technology (M.Tech.) in Communication Engineering Jun 2013 – May 2015
Vellore Institute of Technology *Vellore, TN, India*

Bachelor of Engineering (B.E.) in Electronics Engineering Aug 2007 – Jul 2012
K. J. Somaiya College of Engineering *Mumbai, MH, India*

TECHNICAL SKILLS & CORE COMPETENCIES

- **Programming & Development:** Python, MATLAB, SQL, Git, Linux
- **Machine Learning & AI:** Deep Learning, Computer Vision, Natural Language Processing (NLP), LLMs, SLMs, Action Recognition, Object Detection, Generative AI, Statistical Modeling, Numerical Optimization, Experimental Design, A/B Testing, Hypothesis Testing, Predictive Analytics
- **ML, AI & Data Science Toolkits:** PyTorch, TensorFlow, Hugging Face Transformers, Unsloth, YOLO (Ultralytics), OpenCV, Nvidia TAO, NumPy, SciPy, Scikit-learn, Pandas, Matplotlib, PaddleOCR, ONNX, TensorRT
- **MLOps & ML Model Deployment:** Docker, ClearML, AWS (EC2, S3)